GraphDBManager

System Description

Abstract

This document describes the GraphDBManager support system of the Eclipse Arrowhead framework (documentation edition 5.3.0), as implemented in `ah5-GraphDBManager-java-spring` of the primary Java source tree (`Eclipse-Arrowhead-java`) and deployed with the standalone Compose definition of that folder.

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the GraphDBManager system. GraphDBManager caches and persists SenML (RFC 8428) sensor and actuator data: `graphDataProxy` keeps the latest message per system and service, `graphDataHistorian` the time series. It is a copy of the DataManager support system [1] whose historian is kept as RDF [2] in an Ontotext GraphDB repository [3] instead of a relational database, so the stored data can also be queried with SPARQL [4].

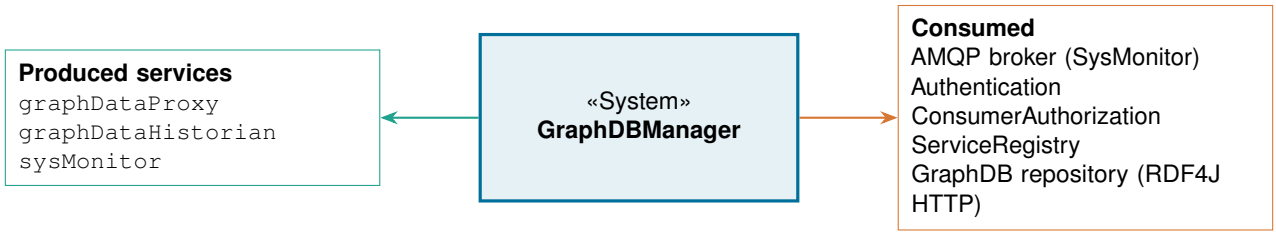


Figure 1: SysML-style block view of GraphDBManager with its produced and consumed services.

The rest of this document is organized as follows. In Section 2.1, we reference major prior art capabilities of the system. In Section 2.2, we describe the intended usage of the system. In Section 2.3, we describe fundamental properties provided by the system. In Section 2.4, we describe delimitations of capabilities of the system. In Section 3, we describe the abstract service functions consumed or produced by the system. In Section 4, we describe the security capabilities of the system.

2.1 Significant Prior Art

The Arrowhead 5.2.2 DataManager [1] (itself a port of the Arrowhead 4.6 DataManager), from which GraphDBManager was copied on 2026-10-28. The API, SenML handling and security are DataManager's; the service definitions are `graphDataProxy` and `graphDataHistorian` because ServiceRegistry reserves `dataProxy` and `dataHistorian` for DataManager. Persistence follows the W3C RDF and SPARQL recommendations [2, 4] as implemented by Ontotext GraphDB Free [3].

2.2 How This System Is Meant to Be Used

Producers `put-data`; consumers `get-data` and browse with `list-systems` and `list-services`. Consumers orchestrate `graphDataProxy` or `graphDataHistorian` through DynamicServiceOrchestration; analytics that need graph queries read the GraphDB repository with SPARQL using a GraphDB account granted by the operator.

2.3 System functionalities and properties

2.3.1 Functional properties of the system

- SenML validation and completion.
- Proxy cache of the last message.
- Historian storage and query by time, count and signals.
- Every stored pack and each of its measurements is an RDF resource in the historian named graph of the GraphDB repository.

2.3.2 Configuration of system properties

The system is configured with Spring properties (`application.properties`, overridden by environment variables in the container deployment). The properties that shape its behaviour are:

Property	Default	Meaning
`server.port`	8467	HTTPS port.
`authentication.policy`	declared	Standalone deployment: `outsourced` with Authentication identity tokens and mutual TLS.
`management.policy` / `management.whitelist`	authorization	Access to the management services.
`enable.blacklist.filter` / `force.blacklist.filter`	false / true	Blacklist check of requesters.
`service.authorization.enabled`	false	Per-call ConsumerAuthorization check (`authorization.verify`, cached 30 s, fail closed); standalone deployment: true.
`normalization.mode`	extended	Name normalization.
`sysmonitor.enabled`, `sysmonitor.broker.*`, `sysmonitor.ssl.*`	true, amqp-broker:5671, server.ssl stores	Native `sysMonitor` publisher of the common library.
`graphdb.url`	http://127.0.0.1:7200	GraphDB server (standalone deployment: `http://graphdbmanager-graphdb:7200`).

Property	Default	Meaning
`graphdb.repository`	graphdbmanager	GraphDB repository id; the repository must exist.
`graphdb.username` / `graphdb.password`	(empty)	GraphDB user with read/write rights on that repository only.
`arrowhead.ssl.crl-file`	(unset)	CRL checked for every client certificate (standalone deployment: `/crl/ca.crl`).

Secrets (GraphDB, keystore and identity passwords, secret keys) have no usable defaults; the standalone deployment reads them from per-deployment Compose file secrets under `/run/secrets`.

The standalone deployment of this folder uses `config/application.properties` (copied into the image) with `server.port`=8467, `authentication.policy=outsourced`, `service.interface.policy=CERT_AUTH`, `management.policy=sysop-only`, `service.authorization.enabled=true`, `mqtt.api.enabled=false`, `arrowhead.ssl.crl-file=/crl/ca.crl`; secrets are read from Compose file secrets under `/run/secrets` (`spring.config.import=configtree:/run/secrets/`). `enable.blacklist.filter` is not enabled there, so incoming requests are not checked against the Blacklist; blacklisted providers are still removed by DynamicServiceOrchestration.

2.3.3 Data stored by the system

Historian SenML packs in the GraphDB repository `graphdbmanager`, named graph `urn:arrowhead:graphdbmanager:historian`, vocabulary `https://arrowhead.eu/ns/graphdbmanager#`; the proxy cache is in memory. Log entries are not stored (no database log appender).

2.3.4 Non functional properties

`echo` is `degraded` while the GraphDB repository cannot be queried.

2.3.5 Stateful or stateless

Stateful: historian records are persistent; the proxy cache is lost at restart.

2.4 Important Delimitations

No retention policy for the historian. No inference: the repository uses the `empty` ruleset. The general Management `get-log` operation has no stored entries.

3 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

3.1 Produced services

Service	Version	Interfaces	SD	IDD
`graphDataProxy`	1.0.0	HTTP(S)	[5]	[6]
`graphDataHistorian`	1.0.0	HTTP(S)	[7]	[8]
`sysMonitor`	1.0.0	AMQP(S)	[9]	[10]

The HTTP interfaces are always offered and the AMQP `sysMonitor` interface with `sysmonitor.enabled=true`.

3.2 Consumed services

Provider	Service	Operations	Use	Condition
ServiceRegistry	systemDiscovery, serviceDiscovery	register, revoke, lookup	Registration of the system and its services; lookup of the bootstrap services.	always
Authentication	identity	identity-login, identity-verify	Own identity token; verification of requester tokens.	`authentication.policy=outsourced`
Consumer Authorization	authorization	verify	Per-call authorization of its services.	`service.authorization.enabled=true`
Consumer Authorization	authorization	verify	Management filter decisions.	`management.policy=authorization`
GraphDB (not an Arrowhead system)	RDF4J HTTP protocol [11]	SPARQL query, transaction	Historian persistence.	always
AMQP broker (SysMonitor)	sysMonitor (publisher)	echo, power, status	Own observations.	`sysmonitor.enabled=true`

Except for the bootstrap services, which are looked up in the ServiceRegistry, the system reaches other systems' services through DynamicServiceOrchestration (`http.collector.mode=ORCHESTRATION`, results cached for 300 s); every such call needs a ConsumerAuthorization grant.

4 Security

The system can run in an insecure mode (`server.ssl.enabled=false`, plain HTTP, identity declared by the requester) for development, and in the Arrowhead secure mode with HTTPS and X.509 certificates following the Arrowhead certificate profile (`<System>.<Cloud>.<Organization>.arrowhead.eu`). Certificates not issued under the cloud CA are refused by the TLS handshake. The standalone deployment of this folder (`config/application.properties` and `compose.yaml`) runs the secure mode.

4.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with mandatory client certificates; MQTT over TLS; AMQP over TLS for `sysMonitor`.
- **Data protection:** transport encryption by TLS; GraphDB is reached over plain HTTP on an internal network joined only by GraphDBManager and its GraphDB, with GraphDB security on (own user, no anonymous access). Secrets are kept out of the container environment (Compose file secrets).
- **System authentication:** by the authentication policy: `certificate`, `outsourced` (identity token verified with Authentication) or `declared` (insecure); the standalone deployment uses `outsourced`. An identity token is accepted only together with the client certificate of the system it was issued to (token-to-certificate binding), and every client certificate is checked against the cloud CA's CRL (fail closed).
- **Authorization of produced services:** management services are protected by the management filter (`management.policy`: `SYSOP_ONLY`, `WHITELIST` or `AUTHORIZATION`); with `service.authorization.enabled=true` ConsumerAuthorization decides each call; an optional blacklist filter refuses blacklisted requesters. Sysop identities bypass the management filter.
- With service authorization every call must come from a client certificate matching the authenticated system, and ConsumerAuthorization must permit the operation.

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

5 References

[1] J. Delsing, "DataManager – System Description," Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-DataManager-java-spring/documentation/v5.3.0/DataManager-v5.3.0-SysD.pdf.

[2] R. Cyganiak, D. Wood, and M. Lanthaler, "RDF 1.1 Concepts and Abstract Syntax," W3C Recommendation, 2014.

[3] Ontotext, "GraphDB 10.8 Documentation," https://graphdb.ontotext.com/documentation/10.8/, 2025.

[4] S. Harris and A. Seaborne, "SPARQL 1.1 Query Language," W3C Recommendation, 2013.

[5] J. Delsing, "graphDataProxy – Service Description (GraphDBManager)," Eclipse Arrowhead, SD 5.3.0, 2026, graphDBManager-v5.3.0-graphDataProxy-SD.pdf.

[6] ——, "graphDataProxy HTTP/TLS/JSON – Interface Design Description (GraphDBManager)," Eclipse Arrowhead, IDD 5.3.0, 2026, graphDBManager-v5.3.0-graphDataProxy-HTTP-IDD.pdf.

[7] ——, "graphDataHistorian – Service Description (GraphDBManager)," Eclipse Arrowhead, SD 5.3.0, 2026, graphDBManager-v5.3.0-graphDataHistorian-SD.pdf.

[8] ——, "graphDataHistorian HTTP/TLS/JSON – Interface Design Description (GraphDBManager)," Eclipse Arrowhead, IDD 5.3.0, 2026, graphDBManager-v5.3.0-graphDataHistorian-HTTP-IDD.pdf.

[9] ——, "sysMonitor – Service Description (GraphDBManager)," Eclipse Arrowhead, SD 5.3.0, 2026, graphDBManager-v5.3.0-sysMonitor-SD.pdf.

[10] ——, “sysMonitor AMQP/TLS/JSON – Interface Design Description (GraphDBManager),” Eclipse Arrowhead, IDD 5.3.0, 2026, graphDBManager-v5.3.0-sysMonitor-AMQP-IDD.pdf.

[11] Eclipse Foundation, “Eclipse RDF4J 5,” https://rdf4j.org/, 2024.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-28	5.2.2	Document created from the DataManager SysD for the GraphDB-backed copy GraphDBManager.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	